

Daily Science Article

Topic: Can You Untangle a Knot?

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The Question

Imagine taking a piece of string, tying it into a complicated knot, and then asking a surprisingly difficult question:

Is there a way to untie it without cutting the string?

At first, this sounds like a practical question. Just pull, twist, and try again.

But mathematicians ask a different version of the question:

How can we prove that two knots are actually the same — or that they are fundamentally different?

This simple question leads to **knot theory**, a branch of topology in which a piece of string becomes a mathematical object and familiar ideas such as “shape” and “same” become much more subtle.

1. When Is a Knot Still the Same Knot?

Suppose we have a closed loop of flexible string.

We can stretch it, bend it, or move it around in three-dimensional space. We are allowed to change its appearance, but we are not allowed to:

- cut the string,
- glue pieces together,
- or let the string pass through itself.

If we can transform one configuration into another using only these kinds of continuous movements, mathematicians consider the two knots **equivalent**.

So a circle and a badly stretched circle are still the same mathematical knot.

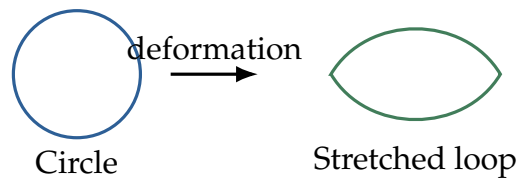


Figure 1. Different-looking loops can represent the same knot under continuous deformation.

This is already a different way of thinking about mathematics.

We are not interested in the exact shape at one moment. We are interested in what remains unchanged while the object is continuously deformed.

The simplest possible knot is called the **unknot**. It is just a loop with no genuine knotting.

The interesting question is what happens when we create a real knot.

2. A Picture of a Knot Is Not the Knot Itself

There is an immediate problem.

A knot lives in three-dimensional space, but drawing a three-dimensional knot on a sheet of paper forces different parts of the string to cross in the picture.

At a crossing, one strand must be marked as passing **over** the other.

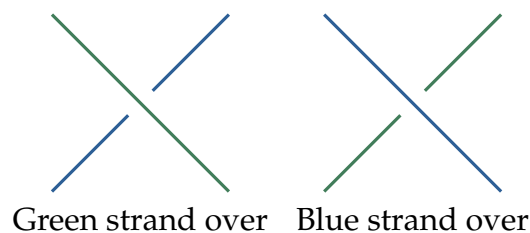


Figure 2. A knot diagram records which strand passes over and which passes underneath at a crossing.

The small gap in the diagram tells us which strand passes underneath, the strand drawn without a break is the one on top.

This means that a knot diagram contains more information than an ordinary drawing. Without knowing which strand passes over the other, we may not even know what knot the picture represents.

Now comes an important discovery.

Even though a knot diagram can change dramatically when we redraw it, there are certain elementary transformations that do **not** change the underlying knot.

These are called the **Reidemeister moves**.

3. Three Moves That Do Not Change the Knot

In the 1920s, Kurt Reidemeister and, independently, Alexander and Briggs established that the three Reidemeister moves provide a complete set of local moves for relating diagrams of equivalent knots.

The first move creates or removes a simple twist.

The second move creates or removes two nearby crossings.

The third move slides one strand past a crossing of two other strands.

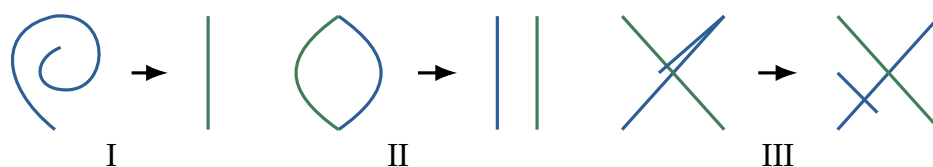


Figure 3. The three Reidemeister moves change a knot diagram without changing the underlying knot.

The remarkable idea is that these moves may change the appearance of the diagram without changing the knot itself.

This gives mathematicians a way to make the statement “these two knots are the same” mathematically precise.

But it creates a new problem.

What if we try many moves and still cannot transform one diagram into another??

Does that prove they are different?

No.

Maybe the sequence of moves we tried was simply not clever enough.

So mathematicians needed something more powerful.

4. The Search for Something That Cannot Change

Imagine that every time you deform a knot, you attach a number to it.

If that number never changes during any allowed deformation, then the number can help us distinguish knots.

Such a quantity is called a **knot invariant**.

If $I(K)$ represents an invariant of a knot K , then equivalent knots must satisfy

$$I(K_1) = I(K_2).$$

Therefore, if we find

$$I(K_1) \neq I(K_2),$$

we immediately know that K_1 and K_2 cannot be equivalent.

This is a powerful strategy:

**Instead of trying to prove two objects are different directly,
find something that one has and the other cannot have.**

5. Counting Crossings

One of the simplest examples is the **crossing number**.

It is the smallest possible number of crossings in any diagram of the knot.

For example, the unknot has crossing number

$$c(\text{unknot}) = 0.$$

A trefoil knot has

$$c(\text{trefoil}) = 3.$$

It cannot be represented by a knot diagram with fewer than three crossings.

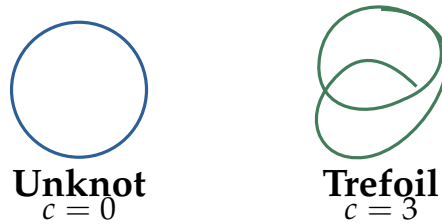


Figure 4. The unknot has crossing number 0, while the trefoil has crossing number 3.

This gives us useful information.

If one knot has crossing number 3 and another has crossing number 4, they cannot be the same knot.

But there is a catch.

Two different knots can have the **same** crossing number.

So crossing number can prove that some knots are different, but it cannot completely identify every knot.

We need something deeper.

6. From Numbers to Polynomials

Here is where knot theory becomes surprisingly algebraic.

James W. Alexander introduced the Alexander polynomial in the 1920s; his work was published in 1928. It became one of the first major polynomial invariants of knots.

Instead of assigning a single number to a knot, we assign it a polynomial.

For the trefoil knot, one common normalization gives

$$\Delta(t) = t^{-1} - 1 + t.$$

For the figure-eight knot, one common normalization gives

$$\Delta(t) = -t^{-1} + 3 - t.$$

These polynomials are different, so the corresponding knots cannot be equivalent.

Strictly speaking, the Alexander polynomial is determined only up to multiplication by $\pm t^n$, so the expressions above represent standard normalizations.

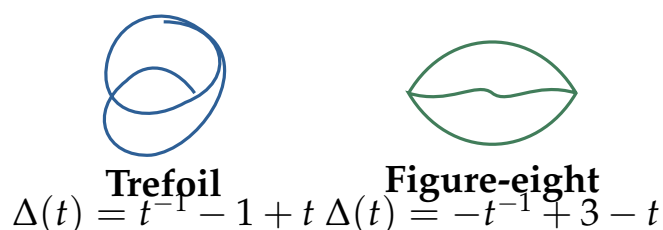


Figure 5. The Alexander polynomial gives an algebraic invariant that distinguishes these two knots.

The important idea is not that everyone should calculate these polynomials by hand.

The important idea is **what the polynomial is doing**.

We have converted a complicated three-dimensional object into an algebraic object.

A knot can look like a tangled piece of string, but mathematics can attach an algebraic fingerprint to it.

7. But Even the Fingerprint Is Not Perfect

At this point, it is tempting to think that the Alexander polynomial solves the entire problem.

It does not.

There are different knots that share the same Alexander polynomial. In particular, there are nontrivial knots whose Alexander polynomial is 1.

So even this sophisticated invariant can sometimes fail to distinguish knots.

And that leads to one of the deeper ideas in knot theory:

There may not be one simple measurement that captures everything about an object.

Instead, mathematicians develop many different invariants.

Some are geometric.

Some are algebraic.

Some come from topology.

Some come from mathematical physics.

And each one reveals a different piece of information.

The search for stronger and stronger invariants became one of the central themes of modern knot theory.

8. Why Should Anyone Care About Knots?

At first, knot theory can sound like mathematics for people who enjoy tying shoelaces.

But its ideas reach much further.

Topology asks questions about properties that survive continuous deformation. Knot theory turns this philosophy into a concrete laboratory: take an object, deform it, and ask what cannot be changed.

That way of thinking appears throughout modern mathematics.

Knot theory, itself a branch of topology, has developed deep connections with algebra, geometry, and mathematical physics.

But perhaps the most interesting lesson is simpler.

When we look at a tangled knot, our eyes immediately focus on its appearance.

Mathematics asks us to look deeper.

Two drawings can look completely different and still represent the same mathematical object.

Two drawings can look almost identical and represent different objects.

And a measurement that seems powerful can still fail to tell the whole story.

9. The Question That Remains

So return to our original question:

Can you untangle a knot?

Sometimes yes.

Sometimes no.

And sometimes the harder question is not how to untangle it, but how to **prove** that two knots are fundamentally different.

That is what makes knot theory so beautiful.

A simple piece of string leads to a mathematical question about **equivalence**.

Equivalence leads to **invariants**.

Invariants lead to **algebraic fingerprints**.

And those fingerprints lead to a much larger question:

How do we recognize what an object really is?

The knot is only the beginning.

The deeper lesson is that mathematics often studies not what changes when we move something, but **what refuses to change**.

And sometimes, discovering what stays the same is much harder — and much more beautiful — than discovering what moves.